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Procurement & Supply Chain Management

13

Operational Excellence, Glencore Copper
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Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

Table of Contents

Preface 2
Table of Contents 3
Figures 3
Tables 3
1. Introduction 4
2. Objectives 4
3. Definitions 6
4. Approach 7
5. Deployment 21
6. Measures of Success 22
7. Improvement 25
8. References 26
9. Revision History 26
10. Appendix A – Stakeholder Signoff 27

Figures

Figure 1-1 Asset Management Framework 4
Figure 4-1 Procurement and Supply Chain Management 8

Tables

Table 4-1 Procurement and Supply Chain Management Process Description 9
Table 4-2 Process RACI 17
Table 4-3 Functional Descriptions 19
Table 6-1 Measure of Success 22
Table 8-1 References 26

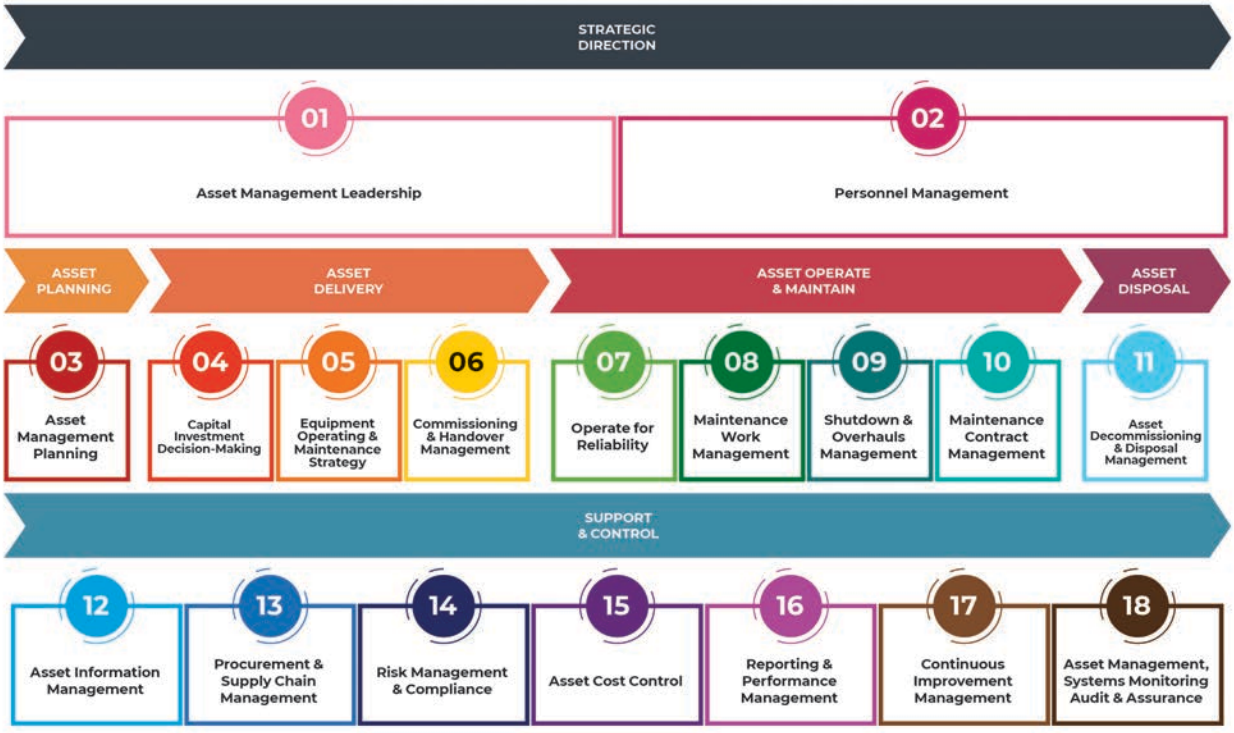
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1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring that assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual is to ensure Procurement and Supply Chain Management processes and systems are aligned with each site's Asset Management Objectives. This objective can be further defined by the following expected outputs:

- The business is not compromised by quality or availability of consumables and spares.
- Decisions on inventory requirements are made in line with the Equipment Operating and Maintenance Strategy to achieve the business goals at the lowest Life Cycle cost.
- Optimised inventory holdings consider newly acquired assets as well as asset decommissioning and disposal.
- Processes are in place for the management of parts and consumables that includes the identification, ordering, issuing and the return of unused items.
- Key indicators that measure the Procurement and Supply Chain performance are developed and implemented.

The effectiveness and efficiency of Procurement and Supply Chain Management can be judged by asking the following questions:

- Does panic buying rarely, if ever occur?
- Are all items unambiguously identified and located?

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- Are all catalogues and lists up to date?
- Are obsolete items promptly discarded?
- Are all items booked out to a work order?
- Are all items inspected before acceptance into stores?
- Are the stores adjacent to the main maintenance workshop?
- Is the store-keeping a full -time job?
- Does maintenance management ultimately decide?
  - What and how much is held?
  - What is obsolete?
  - When sub-assemblies, rather than components, should be held?

If the answers are in the affirmative, it is likely that the Procurement and Supply Chain Management processes and facilities are effective and efficient.

2.1 Accountabilities

Dedicated resources must be assigned and capabilities developed within each Operation to identify work to be able to deliver Procurement and Supply Chain Management processes that enable effective delivery of the Asset Management objectives

Assign accountability for each or all the following:

- Material Specified.
- Material Adequately Stocked.
- Spares Preserved.
- Spares Requisitioned via CMMS.
- Repairables Managed.
- Warranty Items Managed.
- Supply KPI's Agreed.

2.2 Scope

This manual should be read in conjunction with Element E05 Equipment Operating and Maintenance Strategy, E03 Asset Management Planning and E08 Maintenance Work Management.

The scope for Procurement and Supply Chain Management includes:

- Realise fully the benefits of the procurement and supply chain management system by integrating with the computerised maintenance management system (CMMS) and each asset's lifecycle plan.
- Link spares, consumables, energy, water and logistics to the demand requirements of the asset(s) operating and maintenance strategy.
- Involve asset owners in reviewing supply holding levels and locations to balance the operational needs and inventory holding costs.
- Identify spares and materials availability for critical equipment from OEM/suppliers are important in minimising production downtime.
- Understand warehousing and storage procedures are key in preventing spares deteriorating during storage.
- Manage asset inventories in line with Glencore Copper's Purchasing and Supply Standards.

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3. Definitions

Procurement – the process of acquiring goods and services the company needs to carry out its basic requirements. Procurement covers the following list of tasks:

- Purchase planning
- Determining standards of quality
- Identifying suitable suppliers
- Price negotiating
- Financing purchases
- Acquiring required goods and services
- Inventory control and disposal of waste production.

Supply Chain Management - the extensive infrastructure needed to get you those goods. Supply chain management should ultimately be considered one of many responsibilities faced by a procurement function.

Total Cost of Ownership (TCO) - is a calculation method that determines the overall cost of a product or service throughout its life cycle. This method combines both direct and indirect costs.

Multi Criteria Decision Making (MCDM) - is a method used to show or rank a number of limited alternatives by considering several criteria related to alternatives in supplier selection.

Analytical Hierarchy Process (AHP) - is a structured technique for organizing and analysing complex decisions, based on mathematics and psychology.

Inventory - Materials and spares that are catalogued in the CMMS and held by the Supply Department and managed in accordance with relevant procedures in the CMMS. Inventory may also be classed as non-stock in the CMMS and purchased on demand or held by suppliers (vendor managed inventory). All inventory has a stock code and master data catalogue information in the CMMS.

Preference Ranking of Organization Method for Enrichment Evaluation (PROMETHEE) - rather than pointing out a "right" decision, the PROMETHEE method assists decision makers find the alternative that best suits their goal and their understanding of the problem.

Economic Order Quantity (EOQ) - refers to the optimum amount of a spare or consumable that should be ordered at any given point in time, such that the total annual cost of carrying and ordering that item is minimised.

Critical Spare – is defined as a regularly moving consumable or spare which will have a high operational or financial impact if it were to stock out.

Insurance Spare – Materials kept in stock because they can cause the loss of production or result in a safety or environmental issue if not available, however they are not expected to be consumed in the normal operating of the site.

Low Risk – parts within low criticality equipment, should they fail will not cause produce a significant loss event regarding safety, production or the environment.

Non-Stocked Spare – spares with a short lead time that can be procured with minimal impact.

Consumable – identifies all materials that are neither insurance nor critical.

Cataloguing - the process of establishing, maintaining and managing materials data within a company's Enterprise Resource Planning (ERP) system.

Maintenance Repair and Overhaul (MRO) - materials, equipment and supplies used in the production process at a manufacturing plant but are not part of the finished goods being produced.

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Vendor Managed Inventory - Catalogued items in the CMMS that are not held in stock by Glencore Copper, they are held by the supplier under a contractual arrangement. The stock is owned by the supplier until purchased and receipted by Glencore Copper.

Repairable/Rotable Asset – "Repairable/Rotable" spares are parts or assemblies that are carried in the front-line store and which are not automatically re-ordered when they are issued. They rotate through a cycle that includes in-service, to a repair/refurbishment, to inventory and then back to service.

4. Approach

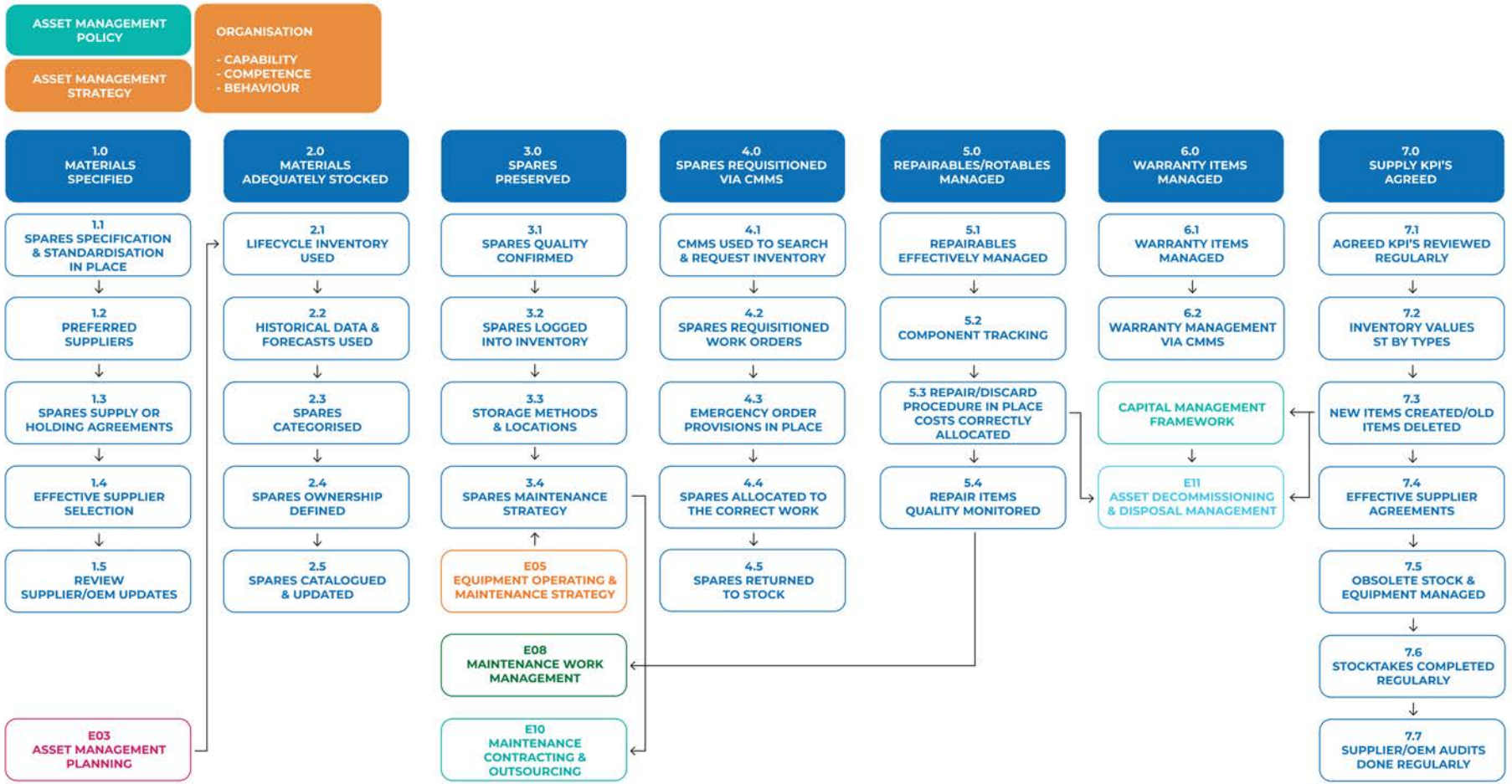
The approach for Procurement and Supply Chain Management is described through:

- Process Flow and Descriptions at Sections 4.1 and 4.2 respectively - illustrates the key Procurement and Supply Chain Management activities.
- Process RACI Matrix - indicates for each activity, the functions who are Responsible, Accountable, who are Consulted prior to final decision/action and those who are Informed of any outcomes and actions, refer to Section 4.3.

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4.1 Process Flow

FIGURE 4-1 PROCUREMENT AND SUPPLY CHAIN MANAGEMENT



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TABLE 4-1 PROCUREMENT AND SUPPLY CHAIN MANAGEMENT PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Material Specified	**Specifying materials to procurement including equipment specifications, consumables and spares details that confirm fit-for-purpose spares supplied to the operation.**
1.1 Spares Specifications and Standardisation in Place	Spare's specifications and brands shall be identified and documented. Spares shall be standardised where practical, this will provide: • Supplier consolidation – provides the opportunity for cost reduction in spares purchasing. • Reduced transactional costs – the purchase of large volumes of spares enables fewer purchasing transactions, therefore reducing transactional costs. • Reduced Total Cost of Ownership (TCO) – specifying spares requirements and brands enables volume purchases of standardised spares therefore reducing the total cost of asset ownership over its life. • Greater control over obsolescence. • Quality Assurance (QA) and Quality Control (QC) are managed appropriately. • The ability to align more effectively to master data standards such as taxonomy, mandated fields etc. • The ability to consistently search and identify spares within the CMMS**.**
1.2 Preferred Suppliers	An evaluation of potential suppliers shall be conducted. Procurement have accountability for achieving best overall value for the Business when acquiring or contracting for goods and services. Procurement is responsible for facilitating the selection process of the best supplier and assuring fair value for the price. It is critical to develop strong relationships with suppliers and go beyond the 'cheapest price' determinant. Procurement to identify preferred suppliers and set up a preferred supplier programme to enable: • Reduced cost – a preferred supplier programme makes it easier to negotiate advantageous pricing and improve payment terms. • Improved quality of service – preferred suppliers better understand the Glencore Copper business due to the more intimate nature of relationship, which means they will be better placed to meet (and exceed) Glencore Copper's expectations. • Communication is more straightforward – when problems arise it's much easier to communicate because there will be a history of working together. • Lower risk – A preferred supplier programme is an important way to mitigate supply chain risk. These suppliers are known, are compliant and will adhere to regulations. An increased level of trust and reliability means that Glencore Copper can benefit in other ways, such as choosing to keep inventory levels low. • Better contractual terms – both Glencore Copper and supplier benefit from a preferred supplier programme. • Increased efficiency – buyers across Glencore Copper will find it quicker and easier to do business with preferred suppliers.

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	• Improved consistency – by definition, preferred suppliers should be as loyal to you as you are to them. They will prioritise Glencore Copper's business and deliver a consistently high-quality service whereas switching suppliers frequently consistent high-quality service may be reduced. • QA and QC to be managed. Hold regular reviews with suppliers and send out internal surveys to measure supplier performance.
1.3 Spares Supply or Holding Agreement	As a minimum the spares strategy should detail the spare parts holding agreements, where Glencore Copper receive a benefit from such agreements, like: • On consignment – where Glencore Copper hold the stock but do not pay until they are required. The supplier owns the stock until Glencore purchases. • Supplier managed – where suppliers manage the ordering of stock on Glencore Copper's behalf and either Glencore hold the stock, or the supplier does. • Standing orders – which stipulates a standard purchase price and quantity. • Forward purchasing agreements – used to lock in a specific price in the agreement period to reduce the risk of price volatility.
1.4 Effective Supplier Selection	Each site is responsible for developing and implementing clear criteria and process for effective supplier selection, examples of such processes are provided below: • Multi Criteria Decision Making (MCDM) • Analytical Hierarchy Process (AHP) • Preference Ranking of Organization Method for Enrichment Evaluation (PROMETHEE).
1.5 Review Supplier/OEM Updates	If any supplier or OEM information or data changes regarding consumables or spare parts, then the recipient shall pass that information to the appropriate person within the supply function who shall update records accordingly.
2.0 Material Adequately Stocked	**Setting in place service requirements, inventory levels and location, and suitable contractual arrangements to enable adequate spares availability to achieve asset management objectives.**
2.1 Lifecycle Inventory Used	Refer to Element E05 Equipment Operating and Equipment Strategy and Element E03 Asset Management Planning. Spare's requirements shall consider the full lifecycle of the asset and be derived from the requirements of Elements E03 and E05. • Determine what spare parts are required for each maintenance type: ○ PM's ○ Corrective (planned and unplanned) ○ Capital Projects. • Determine the criticality of each spare part and establish critical spares list and make reference of the criticality in the CMMS. For example, the ABC indicator in SAP, where A represents the most critical items. Review critical items every 5 years. • Determine lead times for spares to be at site. • Determine the delivery lead time from preferred suppliers.

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	• Inventory is to be checked on a regular basis to ensure correct physical stock holdings are present in correct quantities and locations when compared to CMMS stock on hand. Critical items to be counted semi-annually. • As goods are delivered into the warehouse, a thorough check of the driver's freight manifest is to be conducted by the store person to ensure that the items detailed on the manifest have all been delivered prior to the store person signing and dating the manifest which is often used as evidence of 'Proof of Delivery'. • Items are to be physically inspected to ensure the item is in a suitable condition for storage and use. • End of Life – communicate to supply chain when items are no longer required and supply chain to decide if the item is to be removed from the catalogue. For obsolete items refer to process step 7.5 of this manual.
2.2 Historical Data and Forecasts Used	Determine the spares Reorder Point and Order Quantity. This is determined by: • Historical information regarding spares consumption. • Forecast usage rates derived from E07 Operate for Reliability and E05 Equipment Operating and Maintenance Strategy and in consultation with supply chain. The demand analysis results are to be communicated with Supply to ensure that items are available as per the demand profile. As demand changes the demand forecast shall be updated and communicated to Supply. The demand forecast should inform on both Planned and Unplanned Maintenance activity. • Consideration must be given in the demand analysis process for items that are common between departments and/or sites, to ensure availability of the item. • The goal of the Economic Order Quantity (EOQ) is to balance the competing objectives of carrying less inventory and the costs associated with ordering and holding inventory. Economic Order Quantity (EOQ) to ensure that full value for money is realised.
2.3 Spares Categorised	Allocation of an item to a category is based on factors such as: • Frequency of use and demand forecast as derived by maintenance. • Value, • Potential for repair, • Criticality to the operation, • Current and future availability • Current and future operational requirements for the item. Correct categorisation of inventory items is essential to: • Manage stock holdings, • Correctly report stock holdings • Identification of materials that are no longer required and may be disposed of either by sale to other sites, external sale, or scrap. Spares shall be categorised using the below as a guide: According to the Glencore Copper Inventory policy inventory is categorised as the following: • Business Impact

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	○ A – will have a financial impact if stock out ○ B – will have a financial impact if no work around can be implemented. ○ C – No short-term financial impact. • Inventory Type ○ I - Insurance ○ C- Critical if business impact is A or B ○ Consumable – all other items.
2.4 Spares Ownership Defined	The owner of the spare parts shall be identified. Where practical the sharing or spares between sites should be considered.
2.5 Spares Catalogued and Updated	Spares shall be catalogued as follows: • Identify material for cataloguing. • Submit request for cataloguing. • Review request, verify naming convention and authorise. • Catalogue item. • Add to Bill of Materials (BOM) if required. • Add material to task lists/standard jobs as required. • Maintain the material catalogue, including updating of price. • Align to and be consistent with master data requirements. Data quality excellence is a key enabler for Glencore Copper to achieve its business objectives and is critical in the catalogue data management process. Data quality is the responsibility of all Glencore Copper employees and must be suitable to support: • The ability to consistently search and identify spares within the CMMS. • Purchasing • Inventory control and stock management • BOM's • Contracts. The following should be considered regarding data quality: Completeness: • Catalogue data record is complete and not missing any data values including photo representation of the material item. Validity: • All data values are within specified ranges. Accuracy: • Cataloguer to ensure verification of key data against a trusted external source. For example, make, model and part number with supplier. Consistency: • Cataloguer to ensure prescribed naming conventions for data entry. • Cataloguer to ensure consistent material classification (NSC)

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	• Cataloguer to ensure duplicate material master data records are captured and prevented at the point of entry. Additional duplicate material master reporting to be periodically undertaken to support data consistency. Integrity: • Cataloguer to ensure correct relationships between the data entities – material, catalogue, APL/BOM. Timeliness: • All parties in the process ensure complete information and data is provided in a timely manner.
3.0 Spares Preserved	**Confirming the condition of received, stored and maintained inventory does not influence an asset's operational performance.**
3.1 Spares Quality Confirmed	The material master shall be used to flag when an item requires a technical quality review and that vendor documents such as factory acceptance test certificates are present. The quality of the spare is to be inspected and confirmed on receival of the spare into the store and that all required documents from the supplier/manufacturer are present. The quality check should be a combined Procurement/Supply and Maintenance effort. Quality records are to be developed and retained, which can then be referenced should warranty issues arise. The item is to be returned if the quality does not comply with Glencore Copper quality requirements. NOTE: Hazardous and Dangerous goods are not to be received without the relevant person's authority.
3.2 Spares are logged into inventory	Once the spare has passed the Quality check it is to be catalogue if not already done so. The spare is to be categorised and added to the inventory in the CMMS and attributes populated as per the spare category and criticality. All information accompanying the spare to be checked for correctness and filed accordingly.
3.3 Storage Methods and Locations	All spares and materials must be stored in a manner which ensures that: • Compliance with all applicable standards and regulations is maintained. • Spares and Materials remain in an optimum operable or usable condition, with no determination in performance expected. • Spares and Materials do not create an environmental risk. • Spares and Materials will be readily located and safely accessible when required, particularly for the purpose of in stock maintenance. • Storage bin location referenced in the CMMS.
3.4 Spares Maintenance Strategy	Critical spares that require in storage maintenance shall have a maintenance strategy developed in accordance with Element E05 Equipment Operating and Maintenance strategy and all strategies are to be implemented into the CMMS and E08 Maintenance Works Management Process followed for the routine maintenance of the spare. This ensure that the spare is in optimal condition when it is required to be used and provides evidence of good storage practises should warranty issues arise. If there is a need for specialist contractors to conduct preservation maintenance consult Element E10 Maintenance Contracting and Outsourcing.

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4.0 Spares Requisitioned via CMMS	**Requesting and withdrawing inventory items for maintenance using the CMMS**
4.1 CMMS used to Request Inventory	CMMS is to be used to search for, request or reserve spares for maintenance work.
4.2 Spares Requisitioned Work Orders	All spares for maintenance work shall be requisitioned or reserved through work orders only.
4.3 Emergency Order Provisions in Place	Emergency Purchase Orders shall be used in the event of emergencies that affect the health and wellbeing of Glencore Copper personnel, environmental breach or significant financial loss. Each site is responsible for the development and implementation of an Emergency Purchase Order process and procedure, together with the appropriate authority levels.
4.4 Spares Allocated to Correct Work	Spares that are withdrawn from the inventory shall be allocated to the work order and equipment that created the demand.
4.5 Spares Returned to Stock	All withdrawn spares that are unused shall be returned to the store and re-entered into the CMMS.
5.0 Repairables/Rotables Managed	**Managing repairable items through repair orders and competent asset management personnel making scrap versus repair decisions**
5.1 Repairables/Rotables Effectively Managed	Each site is responsible for the development and deployment of a repairables/rotables process and procedure. Repairables/rotables are to be managed in accordance with that procedure. A typical procedure would consider the following for inclusion: • Identify Rotable • Classify as Repairable or Rotable • Stocking assessment • Rotable item holding levels • Replace Rotable Item Decision Logic • Storage and Preservation • Stocktake requirements • Finance and Cost • Rotable process map and process description.
5.2 Component Tracking	A repairable item work order shall be created (e.g. PM04) and repair details entered, together with the repair scope and factory testing requirements. The repairable item is to be tracked from work order creation to receiving the repaired item back into the store, whether the work has been carried out on or off site. Repairable items shall be classified as equipment in SAP and serialised.
5.3 Repair/Discard Procedure in Place Costs Correctly Allocated	Each site is responsible for developing a procedure that will provide guiding principles regarding the decision to repair or discard an item that has been removed from service. The procedure should also provide information on how costs are going to be allocated in the repairables/rotables process. If an item is to be discarded refer to Element E11 Asset Decommissioning and Disposal Management.

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5.4 Repair Item Quality Monitored	Whether an item is repaired on-site or off-site the quality of the repair is to be monitored at milestone stages of the repair process. A suitably qualified and experienced person is to be sent to the repair facility to inspect the item and assess the process by which the repair is been undertaken and the quality of the repair. Factory acceptance testing should be witnessed. Refer to Element E08 Maintenance Work Management to enable the repair of the item.
6.0 Warranty Items Managed	**Managing warranty items so that they can be identified and claims can be made should the warranty item fail during the warranty period.**
6.1 Warranty Items Managed	Warranty items shall be identified and documented in the CMMS.
6.2 Warranty Management via CMMS	Warranty shall be managed via the CMMS. Should a warranty claim be required a warranty work order is to be raised and the item shall be returned to the supplier, or on-site inspection may be required. Root cause failure analysis to be conducted as part of the post failure investigation and details of the findings passed onto the supplier.
7.0 Supplier KPI's Agreed	**Reviewing agreed supply performance KPIs to confirm time stock delivery, justifying stock levels at site or held by suppliers are acceptable and managing obsolete stock:**
7.1 Agreed KPI's Reviewed Regularly	Refer to Section 7 for supplier KPI's. Review of Supplier KPI's to be conducted annually or subject to supplier performance.
7.2 Inventory Values Set by Type	A balance regarding the amount of Maintenance, Repair and Overhaul (MRO) items in inventory is vital for ensuring that Glencore Copper can deliver on the business and Asset Management Objectives. Too many items in inventory places an unnecessary financial burden on Glencore Copper as holding costs can be as much as 12% of the replacement value of the item per annum. Too few items then prolonged non-planned downtime is a high probability. Inventory value to be split between: • Inactive (obsolete and non-moving) vs Active. • Insurance vs Critical vs Consumable or By Business Impact Code. • Stocked vs non-stocked. Efficient replenishment of inventory is based on a process using information about lead time, demand, costs, criticality and agreed service levels. The primary objective of setting minimum and maximum stockholding levels is to ensure the minimum amount of stock is held to ensure that Glencore Copper operations do not experience stock outs, particularly on critical items.
7.3 New Items Created Old Items Deleted	As new items are brought into the business they are to be catalogued, entered into the inventory in the CMMS and stored in appropriate locations. Conversely if items are removed from service, then they shall be removed from the inventory, the CMMS and working capital adjusted. If suppliers of items change the preferred supplier list shall be updated. Refer to Element E11 Asset Decommissioning and Disposal Management. If a capital item, refer to the Glencore Copper Capital Management Framework.
7.4 Effective Supplier Agreements	Formal supplier contracts and agreements shall be established which should include supplier KPI's and Service Level Agreements (SLA).

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	Things to consider when establishing supplier agreements and establishing supplier relationships, include: • Communication • Respect • Openness • Fairness • Contractual Terms • Value for Money – minimise the total cost of ownership • Mutually Beneficial • Continuous Improvement.
7.5 Obsolete Stock and Equipment Managed	Obsolete spares and equipment place an unnecessary financial burden on Glencore Copper. Obsolete spares and equipment shall be identified, removed from inventory and sold if financially viable, otherwise dispose of according to Glencore Copper policy. A review to identify and manage obsolete stock shall be carried out in real time to ensure that items are management accordingly as situations change. Finance and supply chain shall be consulted and items removed from the catalogue and financial register as required.
7.6 Stocktakes Completed Regularly	Stocktakes are essential to the achievement of two essential inventory management criteria: • Confirmation that the value of inventory is accurately represented in the Glencore Copper financial records. • Confirmation that the inventory management processes are operating accurately and therefore providing confidence that those materials will be physically in stock when the CMMS indicates that they are. Conduct stocktakes on a regular basis; this enables the comparison between the physical stock and stock that has been issued through the CMMS. Discrepancy would suggest lack of adherence to supply process.
7.7 Supplier/OEM Audits Done Regularly	Conduct Supplier/OEM audits on a regular basis. Typically, an audit would consist of: • Reputation • Quality Management (ISO certification) • Document Management and Control • Engineering and Manufacturing processes • House Keeping • Employee qualifications • Communication • Customer Satisfaction • Timing of supply as per the contract.

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4.2 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Procurement and Supply Chain Management process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who ultimately "owns" the activity – there should only be one person.
C	Consulted	Person that needs to feedback and contribute to the activity.
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication).

Process Step \ Function	General Management	Maintenance Management	Operations Management	Operations	Maintenance Execution	Supply / Procurement Management	Supply / Procurement	Maintenance Planning and Scheduling	Finance	Engineering
1.0 Materials Specified										
1.1 Spares Spec. in Place		R	C		C	A	C	C		C
1.2 Preferred Suppliers	I	R	R	I	C	A	R	C		R
1.3 Spares Supply or Holding Agreement	I	R	R		C	A	R	C		R
1.4 Effective Supplier Selection		R	R		C	A	R	C	C	C
1.5 Receive Supplier/OEM Updates		C	C		I	A	R	C	I	C
2.0 Materials Adequately Stocked										
2.1 Lifecycle Inventory Strategy Used	I	A	C		C	R	C	C	C	R
2.2 Historical Data and Forecasts Utilised	I	A	C	C	C	R	C	C	C	C
2.3 Spares Categorised		R	C	I	C	A	R	R	C	R
2.4 Spares Ownership Defined		R	C	I	I	A	C	C	C	C

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Process Step \ Function	General Management	Maintenance Management	Operations Management	Operations	Maintenance Execution	Supply / Procurement Management	Supply / Procurement	Maintenance Planning and Scheduling	Finance	Engineering
2.5 Spares Catalogued and Updated		C	I	I	I	A	R	C	I	I
3.0 Spares Preserved										
3.1 Spares Quality Confirmed		R	I		C	A	R	C	I	C
3.2 Spares Logged into Inventory		R	C	I	I	A	R	C	C	C
3.3 Storage Methods and Locations		R			C	A	R	C		C
3.4 Spares Maintenance Strategies	I	A	C		C	R	C	C	I	R
4.0 Spares Requisitioned via CMMS										
4.1 CMMS Used to Request Inventory		A	R	R	R	R	R	R		R
4.2 Spares Requisitioned Work Orders		A	R	R	R	R	R	R		R
4.3 Emergency Order Provisions in Place	I	R	R		I	A	R	C		R
4.4 Spares Allocated to Correct Work		A	R	R	R	R	R	R		R
4.5 Spares Returned to Stock		A	R	R	R	R	R	R		R
5.0 Repairables/Rotables Managed										
5.1 Repairables Effectively Managed		R			R	A	R	R		R
5.2 Component Tracking		R			C	A	R	R		R
5.3 Repair/Discard Procedure in Place. Costs Correctly Allocated		R			R	A	C	C	R	C
5.4 Repairable Items Quality Monitoring		R			R	A	C	C		R
6.0 Warranty Items Managed										
6.1 Warranty Items Managed		R			C	A	R	C	I	C
6.2 Warranty Management via CMMS		A			I	R		C	I	C
7.0 Supply KPI's Agreed										

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Process Step \ Function	General Management	Maintenance Management	Operations Management	Operations	Maintenance Execution	Supply / Procurement Management	Supply / Procurement	Maintenance Planning and Scheduling	Finance	Engineering
7.1 Agreed KPI's Reviewed Regularly	A	R	C	I	I	R	C	C	I	C
7.2 Inventory Values Set by Type	I	R	I	I	I	A	R	C	C	C
7.3 New Items Created/Old Items Deleted		R	I	I	I	A	R	R	C	C
7.4 Effective Supplier Agreement		C	C	I	I	A	R	C	C	C
7.5 Obsolete Stock and Equipment Managed		R	C	I	C	A	R	C	I	C
7.6 Stock Takes Completed Regularly		C	C	I	I	A	R	C	I	I
7.7 Supplier/OEM Audits Done Regularly		R	R	I	I	A	R	C	I	R

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
General Management	• Aggregate and integrate Asset Management Framework with accountability for its implementation across the site. • Improve organizational culture through effective engagement and motivation of people across the site. • Develop and deliver Life of Asset Plan and Budgets. • Sponsor Site Strategic Asset Management Plan. • Control of maturity evolution across the site.
Maintenance Management	• Identifies critical assets and spare parts. • Identifies which spares parts need to be stocked. • Identifies the maintenance requirements for critical spares while in stores. • Ensures correct stocking strategy is developed and implemented. • Obsolete stock is identified and managed. • Repairables are identified and internal repairs monitored in accordance with Glencore Copper Rotable/Repairable policy. • Warranty items are identified. • Ensure process is followed regarding cataloging of spares, expediting spares and the return of unused spares. • Identifies preferred suppliers. • Provides input into the categorization of spares.

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Maintenance Planning and Scheduling	• Ensures that all spares are identified and catalogued. • Ensures the Bill of Materials (BOM's) are developed. • Ensures that spares are identified for each job and entered into the work order in the CMMS. • Ensures that spares parts are identified and available when required and requisitioned via the CMMS. • Assists in the identification of obsolete spares, warranty and repairable items. • Plans and schedules in stock maintenance. • Schedules jobs that spares are consumed as part of the job. • Assists in the identification of preferred suppliers.
Maintenance Execution	• Assist with checking spares quality. • Identifies spares that need to be returned to the store and returns them. • Ensures that all unused spares are returned to the store. • Ensures that fitting of spares is done in accordance with required procedures and that no rework occurs. • Ensures that the CMMS is used to request spares and through a work order. • Assists in the identification of rotables/repairables. • Provide information to supply regarding warranty items. • Assists in the identification of obsolete parts. • Ensures that in stock maintenance is carried out.
Operations Management	• Identification of preferred suppliers in the identification of suppliers. • Ensures that spares are requisitioned vis the CMMS
Supply/Procurement Management	• Ensures that the quality of the spares received at the warehouse are to Glencore Copper Quality Standards. • Ensures that supplier selection is done in accordance with Glencore Copper Supply Chain and Procurement Policy. • Ensure that there are adequate spare parts available and that they are stored in appropriate locations to ensure they are protected while in stores. • Allows entry into the warehouse/store to maintenance personnel carrying out in stock maintenance. • Manages the rotable/repairable process and items. • Warranty items managed. • Ensures that warranty claims are progressed. • Ensures that stocktakes are performed in accordance with Glencore Copper standard. • Obsolete spares are removed from stock. • Ensures that vendor managed inventory is available when required. • Ensure that external and internal repairable items are managed in accordance with Glencore Copper Rotable/Repairable policy.
Supply/Procurement	• Assists to ensure that the quality of the spares received at the warehouse are to Glencore Copper Quality Standards. • Assists to ensure that supplier selection is done in accordance with Glencore Copper Supply Chain and Procurement Policy. • Ensure that there are adequate spare parts available and that they are stored in appropriate locations to ensure they are protected while in stores. • Allows entry into the warehouse/store to maintenance personnel carrying out in stock maintenance. • Assists to ensure that warranty claims are progressed. • Ensures that stocktakes are performed in accordance with Glencore Copper standard. • Obsolete spares are removed from stock. • Ensures that vendor managed inventory is available when required. • Assists to ensure that external and internal repairable items are managed in accordance with Glencore Copper Rotable/Repairable policy. • Ensures that spares are managed to ensure optimal condition and performance when needed.
Finance	• Provide input into the management of obsolete items. • Provides advise on general financial matter regarding supply and procurement

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Engineering	• Assists in the identification of critical spares. • Assists in the identification of the quantity of spares parts to hold. • Assists in the identification of vendor managed inventory. • Ensures that the specification of spares meets the needs of Glencore Copper. • Ensures the quality of repairable/rotable items. • Develops the in-stock spares maintenance strategy and ensures historical data is kept and of good quality. • Assists to ensure that supplier selection is done in accordance with Glencore Copper Supply Chain and Procurement Policy.

5. Deployment

The deployment of the Procurement and Supply Chain Management approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching and mentoring.

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies to:

- Guide recruitment, selection, development and deployment of people in asset management roles across the company.
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

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5.4 Training, Coaching and Mentoring

Coaching and mentoring are another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Measures of Success

Refer to Glencore Copper's AM Manual E16 Reporting and Performance Management Manual for current KPI's.

NOTE: *Calculation methods and performance indicators will be updated in line with the completion of the SAP Materials Management (SAP MM) metrics by the procurement and supply chain team.*

TABLE 6-1 MEASURE OF SUCCESS

Metric	Stock Health Status
Definition	Inventory Health Status: • Surplus Order • Good State • Potential Stockout • Stockout.
Performance Indicator	• Surplus Order – Net stock not above the max level • Good State – Stock level between min and max • Potential Stockout – Net Stock on hand below min level with no open order • Stock Out – Demand > Net Stock.
Calculation	See note above
Rationale	• Surplus Order – Defined as items that have Net Stock above the maximum level but is not defined as excess due to open orders. • Good State – Defined as items having net stock between minimum and maximum level but is not defined as excess due to open orders • Potential Stockout – Defined as items that are not completely stocked out but is at risk to be stocked out soon due to the Net Stock on Hand being less than the minimum level and there is no open order. • Stockout – Defined as items that have a higher demand as its current Net Stock on Hand.
Action	• Surplus Stock – No more open orders until min level triggers new order • Good State - no action required • Potential Stock Out – ensure that auto order is activated • Stock Out – review min/max levels and adjust as required. Ensure auto order is activated.

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Metric	Materials Management Master Data Accuracy
Definition	Master Data accuracy of changes made to materials that have been reserved/issued (active) each month.
Performance Indicator	% of changes made on active materials per month and trending down.
Calculation	Number of change requests received for the month/Number of active materials.
Rationale	Master data improvement is an ongoing process and as such there needs to be a mechanism by which data improvement can be measured. The intent of the measure is to understand how many materials are modified against the number of materials that are being used (active) on a consistent basis.
Action	Requests are actioned in a timely manner.

Metric	Stock Availability
Definition	Measures the % of days availability of stock and on demand items. Business impact codes A, B and C impacts the target.
Performance Indicator	Impact Code: A – 99% availability B – 95% availability C – 80% availability Tracked daily reported weekly
Calculation	Total days stock available/Total days in reporting period.
Rationale	Stock availability provides assurance that items will be available when called upon. Stock is classified into three main groups based on the importance (criticality) to the organisation using A, B, C coding system, with items A being the most important.
Action	If stock is not available when a demand is requested carry out the following: • Expedite stock to ensure availability • Demand analysis • Min/Max analysis • Lead time analysis • Ensure impact code is correct.

Metric	Procurement Savings Realisation
Definition	Identify and report on the value of material and procurement savings per category, across the sourcing hub five (5) pillars: Bulk and logistics, Mining, Fixed Plant, HME and Services.
Performance Indicator	% Savings per month for materials % Savings per month for services
Calculation	Cost of materials this month/Cost of materials last month

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	Cost of services this month/Cost of services last month
Rationale	Negotiation of material and services can yield significant cost benefit While cost is an important aspect of procurement savings, consideration should be given to other factors such as, quality, quantity and timeline. Cost savings should not be at the detriment of these factors.
Action	If no cost benefits have been realised, then consideration of alternative suppliers should be given.

Metric	**Purchase Requisition (PR) to Purchase Order (PO) Performance**
Definition	Measures the cycle time from when a PR is submitted to the time when it is transmitted (PO) to a vendor or contractor, for inventory managed items.
Performance Indicator	PR to PO < 2 days.
Calculation	See note above
Rationale	Measures Glencore Copper internal process capability to ensure that Purchase Orders are transmitted to vendors and contractors in a defined period of time. This ensures quick turnaround time for goods and services, improve staff productivity and reduce supply and procurement function cost.
Action	Investigate the reason for cycle time deficiency, instigate improvement actions and implement.

Metric	**Supplier DIFOT**
Definition	Measure of goods/services delivered within the stated lead times.
Performance Indicator	95% of all items delivered in full and on time.
Calculation	Number of deliveries in full and on time/Number of deliveries requested.
Rationale	It is important that materials and services are delivered in full and on time so as not to impact on plant and equipment availability and ability to meet production targets. It is a measure of supplier delivery performance with regards to: • Product/service quality • Correct quantity • Correct place of delivery • At the time requested.
Action	Discuss performance with vendor/supplier, if performance deficiencies continue to exist then consideration for a change in vendor/supplier should be given.

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Metric	Stock Accuracy
Definition	Measures the number of stock items on hand quantity with the number of items showing in CMMS/Optimiser?
Performance Indicator	<5% discrepancy
Calculation	Number of Stock items on hand/Number of Stock in CMMS.
Rationale	Stock accuracy measurement is important as it provides information regarding a number of items: • Whether too much stock is being carried – increase in stock cost • Whether to little stock is being carried – potential for stock availability impact • Master data accuracy – to many or too few records – could increase the cost of maintaining data or cause potential stock outs.
Action	The following actions should be considered: • Stock carried is above required level – remove surplus stock and adjust master data as required • Stock carried is below required level – investigate why, adjust carrying levels and adjust master data as required • CMMS has to many items on record – adjust master data • CMMS has insufficient items – adjust master data.

Metric	Goods Receipt Performance
Definition	Measures how long it takes to process goods from receipt of the goods until the goods are available in stock and ready for issue.
Performance Indicator	< 1 Days
Calculation	See note above.
Rationale	To ensure that goods are available when required it is important that they are available for issue when required and as such there needs to be a measure in place to understand internal process performance for the receival of goods to the goods being available in stock for issue.
Action	Assess internal process to identify process gaps, put actions plans in place to close identified gaps.

7. Improvement

The process to review and continually improve the approach and deployment for Procurement and Supply Chain Management is scheduled Annually, or when there is a sudden change in the business strategic and operational requirements. As the organization matures a longer review period could be considered.

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8. References

The following documents provide detailed support material that should be read in conjunction with this Standard to provide a comprehensive understanding of Asset Information Management.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Asset Management Metrics Standard	This document defines the measures/KPIs, the method of calculation, the intent and behaviours that each measure is aimed at.
Purchased Goods and Inventory Management Protocol	Glencore Coal protocol regarding: • Receipt and issue of direct purchased materials • Storage of inventory in a manner that contributes to operational reliability and cost minimisation.
Rotables Management Procedure	Glencore Copper Mount Isa Mine Rotables Management Procedure.
Procedure for Review and Setting of Min/Max	Glencore Copper Mount Isa Mine Stock Min/Max Review and Setting Procedure.
Cataloguing of Material Master Data Procedure	Glencore Copper Mount Isa Mine Cataloguing of Materials Master Data.

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	28/04/2021
0.2	Gather Documents and Create Draft Manual	10/06/2021
0.3	Review and Signoff 1st Draft Manual with Sites	11/06/2021
0.4	Reformat and Issue for Final Review	31/08/2021
1.0	Update Site Signoff and Publish	20/05/2022

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10. Appendix A – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	7/6/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	7/6/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	8/6/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	10/6/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	7/6/2022
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Procurement & Supply Chain Management

13

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